

Multiscale LSTM-Based Deep Learning for Very-Short-Term Photovoltaic Power Generation Forecasting in Smart City Energy Management

Dohyun Kim¹, Dohyun Kwon, Laihyuk Park, Joongheon Kim², *Senior Member, IEEE*, and Sungrae Cho³

Abstract—Photovoltaic power generation forecasting (PVGF) is an attractive research topic for efficient energy management in smart city. In addition, the long short-term memory recurrent neural network (LSTM/RNN) has been actively utilized for predicting various time series tasks in recent years due to its outstanding ability to learn the feature of sequential time-series data. Although the existing forecasting models were obtained from learning the sequential PVGF data, it is observed that irregular factors made adverse effects on the forecasting results of very-short-term PVGF tasks, thus, the entire forecasting performance was deteriorated. In this regard, multiscale LSTM-based deep learning which is capable for forecasting very-short-term PVGF is proposed for efficient management. The model concatenates on two different scaled LSTM modules to overcome the deterioration that is originated from the irregular factors. Lastly, experimental results present the proposed framework can assist to forecast the tendency of PVGF amount steadily.

Index Terms—Deep learning, long short-term memory (LSTM), photovoltaic power generation prediction, renewable energy.

I. INTRODUCTION

RENEWABLE energy resources such as PV, wind, and geothermal power resources are spotlighted by numerous global agreements, which emphasize the importance of eco-friendly energy generation [1], [2]. Such green energy resources are expected to cover the ever increasing demand of various applications in smart city such as charging stations for electric vehicles (EVs) and domestic service as Fig. 1. In addition,

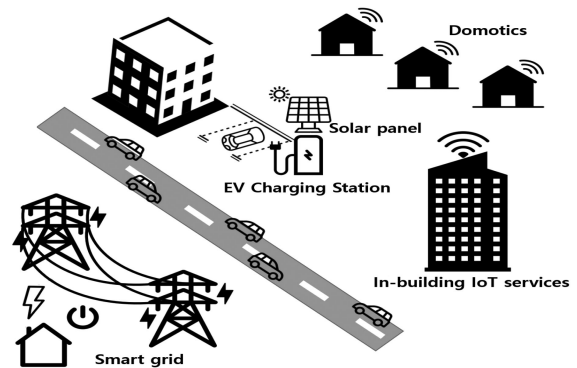


Fig. 1. Electricity usage scenario of smart city. The generated PV power can be supplied to meet the demand of EV vehicle charging station.

they can also significantly contribute to the amount of total energy supply of existing electrical systems [3]. Specifically, the PV power is considered to be attractive because no extra cost is required except initial installation costs to operate the generation plants. However, because the amount of PV power is significantly influenced by weather condition, efficient and accurate PVGF is crucial for managing electricity supply in smart city.

The amount of PVGF is affected by the irregular factor such as *solar ramping* [4]. Such PV-related meteorological factor is hard to be modeled with previous probabilistic approaches and, thus, causes the fluctuation of the amount of PV power generation, so that it makes the PVGF task more challenging. Therefore, recent PVGF research was highly motivated to improve to accomplish more accurate estimation of electricity generation to mitigate the outage phenomenon. The research results are based on Markov chain [5], probabilistic model such as support vector regression (SVR) [6], [7], and artificial neural networks (ANN) [8], [9].

Although the basic ANN and probabilistic approaches were able to forecast the PV power generation, they had limitations which did not reflect the features of time-series data. In addition, they disregarded to learn independent features derived from training data without considering the time domain issue. To overcome this limitation, the recurrent neural network (RNN) and long short-term memory RNN (LSTM RNN), which improved *vanishing gradient problem* of traditional RNN, were suggested as alternative ways for time-varying PVGF problems and were broadly adopted for training sequential time-series PVGF data.

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Motivation: The meteorological event such as solar ramping is hard to be modeled because of its unpredictable feature. Usually, the solar ramping mostly happen unexpectedly and temporarily (e.g., imagine a situation where a piece of cloud blinds PV power generation panels). In this article, this feature is called as *unexpected and unpatternizable intrinsic temporal* (UIT) feature. However, most previous research focused on long-term PV forecasting tasks and neglected time series features of data so that adaptable electricity management with previous approaches for smart city was hard to be conducted. Thus, we propose a forecasting model, which is not only based on LSTM RNN to reflect the correlation between time series PV generation data, but hourly forecasts the PV power generation to robust on solar ramping issue. The advancements were highly focused on effectively extracting intrinsic features of solar ramping of time-series data, and stabilizing the training process of forecasting model. That is, the primary motivation for utilizing the LSTM RNN as a baseline of the proposed forecasting model is the ability to model the feature of solar ramping, which lies on the time series PVGF data. In addition, because the tendency of similarity between ground truth and forecasting result is important factor, the proposed model aims to forecast the PV generation as similar to the variation of ground truth. In this article, such ability which follows up the ground-truth is called reactivity. Overall, the proposed hourly very-short-term (VST) LSTM RNN-based PVGF model aims to be robust on such UIT feature and consider time series features to improve the efficient electricity management of smart city.

Main objective: The objective of this article lies on following improvements: this article aims to improve the previously proposed architecture of the LSTM RNN forecasting model to enhance reactivity regarding UIT-features-related issue while enhancing the forecasting performance of the existing LSTM RNN model. Note that the objective of this article is not focused on designing a PV power forecasting model, which outperforms previously proposed state-of-the-art (SOTA) methods, but confirming feasibility of improvement of the forecasting model by enhancing the reactivity. For this reason, the standard architecture of the LSTM RNN model is adopted as the baseline model. To this end, the multiscale LSTM-based deep learning framework for VST PVGF, which consists of two different joint LSTM RNN models, is proposed. One of the LSTM RNN models is considered for extracting the macroscopic features related to PV power forecasting, while another one is for reducing microscopic features of the solar ramping. That is, the only difference of these models is the duration of time series input data fed into each model. Henceforth, the model with shorter input is called as *short LSTM* and the other one is called *long LSTM*. The experiments to evaluate the forecasting performance of our proposed model were conducted with data collected from three different areas. By training the data, the proposed multiscale LSTM VST-PVGF model aims to show the accurate and reactive forecasting performance than previous baseline LSTM RNN PVGF model.

Contribution: The main contributions of this article can be summarized as follows.

- 1) Based on the motivation which UIT features affect the performance of VST-PVGF task, attempt to improve the

reactivity of the proposed model with applying the UIT features is conducted. That is, multiscale LSTM-based deep learning framework, which jointly concatenates bottleneck features derived from both short LSTM and long LSTM PV power forecasting models is proposed.

- 2) Data-intensive evaluations are conducted with Python/TensorFlow-based software using real-world data obtained from three different places in South Korea and forecasting models trained from aforementioned real data are evaluated with various metrics including root mean square error (RMSE), symmetric mean absolute percentage error (sMAPE), and newly proposed mean absolute derivative error (MADE).
- 3) Throughout various experiments, the performance of the proposed LSTM RNN PV forecasting model (multiscale LSTM-based deep learning framework) is verified and is available to improve the base line forecasting model.

The rest of this article is organized as follows. Section II introduces related recent research on PVGF and basic architecture of the LSTM RNN model. Section III defines the problem in VST-PVGF and presents the details of the proposed model. In Section IV, the evaluation results are presented. Finally, Section V concludes this article.

II. RELATED WORK

As mentioned before, the PV power generation is expected to take a key role of energy infrastructure of smart city. In this regard, recent research focused on an optimization of PV-related problem such as economic deployment of the energy storage system in PV-integrated EV charging stations of smart city [10]. In addition, PV power generation predicting models were studied in [11] and [12] for various industrial applications of smart city. In this section, related various PVGF research and basic LSTM RNN architecture are discussed.

A. Direct and Indirect Forecasting

The PVGF research directions include *direct* and *indirect* approaches. For the direct approaches, the amount of PV power generation is estimated by utilizing the history of generated PV power. On the other hand, the indirect approaches estimate PV-related meteorological data and then approximate the amount of PV power generation. Based on the approaches, ANN-based methods [8], [9], [13]–[15] and non-ANN approaches [20], [27] have been proposed. Nearly all previous methods focused on accurate forecasting of photovoltaic power generation for various time scales. However, they were limited to conduct the VST-PVGF task because they did not consider the solar ramping related feature (i.e., UIT feature), which is highly related to the reactivity of forecasting model.

1) Direct Approach: The direct approach forecasts the PV output by analyzing input data, including meteorological data and/or the history of PV power generation data for sequential spatial/temporal information. The authors of [20] proposed probabilistic PVGF approach, which relies on an extreme learning machine (ELM) as a fast regression model. The ELM is trained in varied ways to obtain quantile forecasts of solar power generation. The method in [21] considered three PV

output power forecasting approaches with ANN, regression, and Schokley-Sandia-based models. The forecasting performances are evaluated with normalized root mean square error (nRMSE). In addition, the methods in [14] and [15] exploited the Levenberg–Marquardt algorithm with multilayer perceptron (MLP) for PVGF.

2) *Indirect Approach*: On the contrary, the indirect approach predicts the amount of solar irradiance and then forecasts the expected PV power generation with the predicted solar irradiance, which dominantly impacts PV power generation. An hourly, daily, and monthly solar irradiance prediction model is proposed in [24], which is highly related to PV power generation. The method was used to design three different models including linear least squares regression, classical backpropagation ANN, and LSTM. A solar irradiance forecasting with the LSTM model was proposed in [25] and it designed feedforward ANN, SVR, and LSTM based prediction models with different weather conditions, i.e., scattered clouds, clear-sky, and overcast. In [26], the method predicted solar irradiance to indirectly forecast PV power generation with combinational ANN approaches. On the other hand, non-ANN approaches are presented in [27] for forecasting solar irradiance. The approach in [27] applied wavelet transform to inputs including historical solar irradiance and decomposed them into several corresponding distinct time-frequency areas. In addition, principal component analysis is performed to extract distinguished inputs, which affects solar irradiance conditionally for each time-frequency area.

B. LSTM Recurrent Neural Networks

Among the aforementioned prediction models, recently developed LSTM-based RNN deep learning methods have shown remarkable results [28]–[30]. Besides solar irradiance or PV power generation problems, time-series-data-related prediction tasks such as wind power prediction in [31] was more successfully conducted by LSTM-based approaches.

The RNN-based model recursively feeds the previous results of input data to itself. Although the forecasting performance of RNN trained by a few sequential input data is remarkable, the performance is deteriorated if the number of input data duration increases. In other words, the RNN model with long duration of input data to reflect the macroscopic history of power generation can deteriorate the overall performance of itself. This issue is caused by the vanishing gradient problem, which is one of the major hurdles for learning long-range dependency. The issue limits RNNs to define the dependency among the information carried by two inputs, which are separated by a long duration.

To tackle the problem, the LSTM-based models are used and they are superior to learn time-series data due to their structural compositions including the forget gate, input gate, and output gate. Among the three gates, the forget gate of the LSTM controls how much the previous results affect the current learning procedure. This characteristic of LSTM addresses the vanishing gradient problem so that the initial data can affect long-range. In other words, as an extension to RNN, the LSTM is capable of learning long-term dependencies by replacing the simple activation nodes with specially designed blocks, each containing cell, input gate, forget gate, and output gate. Consequently,

the LSTM-based RNN is naturally capable of learning both long-term and short-term dependencies of the input sequences. This property makes LSTM suitable for PVGF tasks where both short-term and long-term (i.e., multiscale) contexts should be jointly considered.

III. MULTISCALE LSTM DEEP LEARNING FRAMEWORK

PV output power forecasting can be divided into several categories based on the time horizon: VST/short-term, mid-term, and long-term [19]. In this section, the overall structure of proposed multiscale LSTM-based deep learning framework for VST-PVGF is introduced. At first, the properties of VST-PVGF are defined. In addition, the limitation to previous LSTM-based models is analyzed. Lastly, the congruence of the proposed multiscale LSTM-based model is presented with two key features, which affect PVGF.

A. Properties of VST Forecasting

Aforementioned neural network including ANN, RNN, and LSTM RNN commonly derive generalized forecasting model by learning intrinsic feature from given training data. Thus, the performance of forecasting with respect to structure of model may vary, but the models fundamentally aim to learn features of power generation from given data. However, in realistic PVGF, there exists factors that are difficult or unable to be modeled such as sudden rainfall, unexpected PV panel malfunction, or migration of scattered clouds. Considering one of these examples, in case of clouds, situation where suddenly appeared clouds eclipse the PV panel can be envisioned. The cloud is considered that it clearly affects to the amount of PV power generation, but it is hard to observe the occurrence of cloud. Besides the cloud, there exists factors that affects the amount of PV power generation and they are also hard to be observed. Although these factors could be observed, the variation of them is another consideration for forecasting PV power generation. Note that this assumption is out of scope.

Mostly, the aforementioned property derived from the unexpected situations is considered as noise so that it could be ignored or can be eliminated in the preprocessing procedure. Since the effects of the factors are mitigated after a few hours in general, thereby being diluted in middle-term, long-term PVGF task. On the other hand, these unexpected elements are not negligible in the VST-PVGF task, even if they occurred for a moment, because it focuses on hourly power forecasting rather than the average of the sum of a few hours' forecasting.

B. Single-Model LSTM for VST-PVGF

Compared to other ANN models such as MLP and convolutional neural network (CNN), the proposed LSTM-based deep learning framework can process longer-duration data as input and can reflect time-series-properties due to its unique forget cell and recursive neural structures. This article considered the LSTM RNN architecture as current research movement. As aforementioned, the objective of this article is not focused on designing a PVGF model, which outperforms previously proposed SOTA methods, but confirming feasibility of improvement of

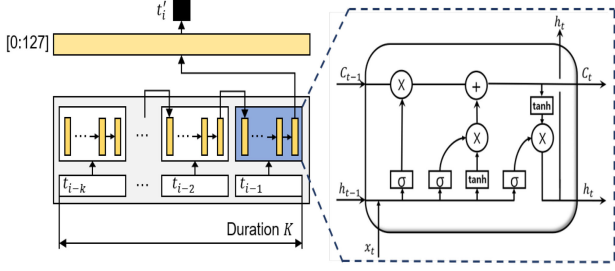


Fig. 2. Single model-based LSTM prediction model and structural composition of an LSTM cell.

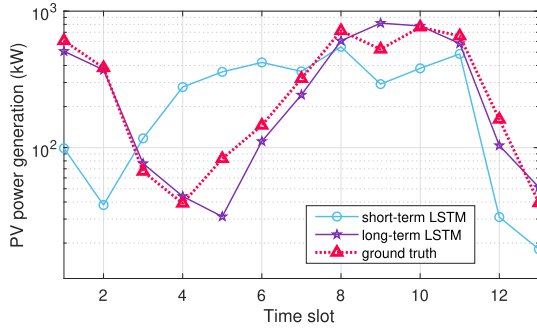


Fig. 3. Feature of short-term and long-term LSTM models: *reactivity* and *accuracy*. The forecasting results is evaluated with manually selected events when rain shower occurred.

the forecasting model by considering the UUIT features for the proposed forecasting model. Thus, for analyzing and evaluating our proposed forecasting model, single model-based LSTM RNN-based forecasting model was utilized as baseline. In turn, it can effectively extract long-term characteristics inherent from the data. However, as the duration of the input data considered in one prediction becomes longer, the importance with the recent data becomes blurred and vanished so that the procedure of training the forecasting model cannot be progressed. That is, if the duration K in Fig. 2 is getting longer, the vanishing gradient problem is highly occurred. Note that the x_t , h_t , and C_t in Fig. 2 stands for the input data, hidden state, and cell state, respectively. In addition, the ρ and $\tanh$ in Fig. 2 stands for the sigmoid and hyperbolic tangent function. By keeping change the h_t and C_t , the LSTM cell can learn long sequence data. Based on the fact, an intuitive way to avoid the vanishing gradient problem is to design the LSTM-based model to use a shorter duration such as 6 h, rather than a longer duration such as 24×7 h. Then, the relation with the immediately preceding data can be emphasized, whereas the ability to extract the long-term pattern becomes weak, since the scale of the input data considered in one prediction becomes small. This is accompanied by a significant reduction in the prediction performance.

Besides, the ability of forecasting model to promptly adapt its result similar to the ground truth is important. If this ability of forecasting model is good enough, then it stands for that the model is robust on solar ramping. In Fig. 3, the relation of reactivity and accuracy over the duration of the LSTM models is presented with forecasting results of manually selected events that rain shower occurred locally during a few hours

(between time slot 8–10). In this case, temporal rain shower is regarded as UUIT factor since baseline LSTM RNN model used in this article have no ability to forecast the occurrence of UUIT factor. Although there may be a high-performance model, which is able to forecast locally occurred rain shower that is totally different scope from this article. Due to the UUIT factor, temporal variation can be found as the red lined between time slot 8 and 10. However, as the purple-lined results, long-term LSTM does not reflect this variation. On the other hand, short-term LSTM seems to reflect this variation. However, the capacity of short-term LSTM model is not sufficient to have forecasting ability so that unpleasant difference can be resulted as the blue-line between time slot 2 and 6, and it also can be even for UUIT factors. Table I shows the result of performance evaluation while proportionally increasing the duration (6 h–1 week). As aforementioned, it seems longer LSTM [model (f)] is prone to result higher accuracy. Interestingly, the model that has medium-length duration [model (c)] results lower performance than the shortest-term model. This can be seemed due to the fact that the medium-length model has no advantage in terms of reactivity as well as capacity.

Thus, for remaining the forecasting performance of the proposed model, the duration of input data should be maintained and weights that represent the latest occurred event should be increased. For accomplishing this principle, it is worth to consider that duplicating some of the latest input data to the end of the input vector. This process is roughly as same as double the weights of entered the latest input data. However, in case of UUIT factors are not included in the latest event, this may cause the model to be biased the latest event so that overall performance of the forecasting LSTM RNN model can be deteriorated.

C. Multiscale LSTM-Based Deep Learning for VST-PVGF

The overall architecture of the proposed framework is illustrated in Fig. 4. Given the time-series historical PV and meteorological data, two different time-scale of data (6-h duration and 1-week duration) are fed into short and long LSTMs. Each network is oriented to the output-specific features of microscopic meteorological and macroscopic PV power generation data with ratios of 32 and 96. Finally, the extracted features are concatenated and are utilized for the PVGF of the next hour after feedforwarding through the fully connected (FC) layer.

In this context, two types of objectives of this article can be summarized, which should be jointly considered: 1) re-emphasizing recent data by appropriately reflecting UUIT factors and 2) maintaining the overall forecasting accuracy of the baseline LSTM/RNN model. Finally, multiscale LSTM-based deep learning framework, which combines the bottleneck features of *short-term* and *long-term* LSTM networks, is proposed to exploit both the advantages of short-term LSTM, which can re-emphasize UUIT factors with weighting intrinsically and the forecasting ability of long-term one.

The bottleneck features [32], [33] are the input values of the latter layer and can be understood as the result of the embedding of the higher dimension input data used for prediction. At the former layer, general domain features are extracted and at the latter layer, task-specific features are trained. This is based

TABLE I
PERFORMANCE OF THE PREDICTION MODELS OVER THE DURATION VARIATIONS OF INPUT DATA (DATA GATHERED FROM YEONSEONG)

Results of each experiment	Epoch	RMSE	sMAPE
(a) single model-based LSTM (duration: 6 h)	1218	15.343	51.941
(b) single model-based LSTM (duration: 12 h)	1171	15.451	52.063
(c) single model-based LSTM (duration: 24 h)	1020	15.696	49.181
(d) single model-based LSTM (duration: 24×2 h)	1029	15.521	46.053
(e) single model-based LSTM (duration: 24×4 h)	1126	14.310	44.992
(f) single model-based LSTM (duration: 24×7 h)	1148	14.054	44.558

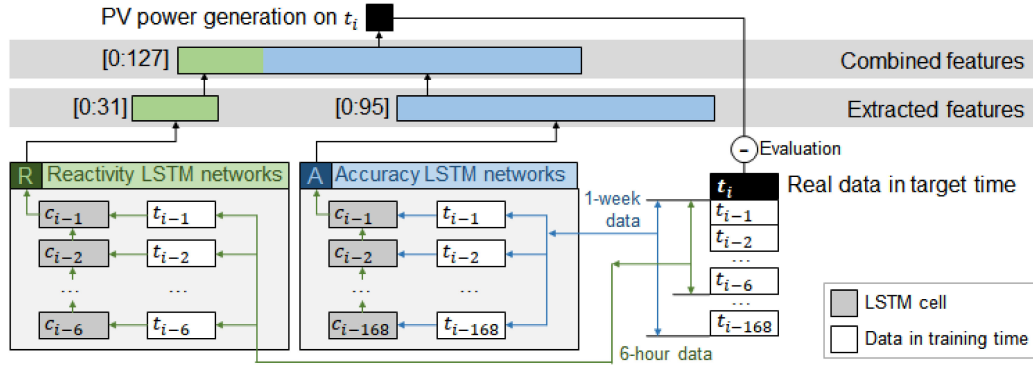


Fig. 4. Proposed multiscale LSTM-based deep learning framework for VST-PVGF.

on the background property of transfer learning [34], [35] that freezes weights of the pretrained model for tasks in different domains and fine-tunes the FC layer for the target task. In the proposed LSTM-based deep learning framework, the bottleneck features of the short-term and long-term LSTM models are the outputs from the last LSTM cell (or the inputs of FC-layer). According to the mentioned characteristics, the bottleneck features in the short-term LSTM model show the embedding result and emphasize the relation with the recent data with weighting according to trained weights. On the other hand, the long-term LSTM model shows the embedding result in the long-term patterns, such as the periodicity. Therefore, the proposed multiscale LSTM-based deep learning framework can reflect both the reactivity and the accuracy with considering these two concatenated embedding results.

IV. IMPLEMENTATION AND EXPERIMENTS

A. Software Prototype Implementation

Each component in the proposed multiscale LSTM-based deep learning framework was implemented with TensorFlow deep learning library. We used a multi-GPU platform (equipped with two NVIDIA Titan V GPUs that have 24 GB memory in total) was utilized for training the proposed neural network framework. Note that after finishing the learning process, testing the performance of model can be processed on CPU-only environment. To ensure fair comparisons, every model has the same architecture with 128 LSTM hidden units, 128 dimension bottleneck features (FC layer input dimension), and an FC layer. The duration of short-term and long-term LSTM models is set to 6 and 24×7 h (a week), respectively. The duration of input data and the number of LSTM hidden units are set identically.

The bottleneck features of the multiscale LSTM-based model are concatenated with 32 and 96 ratio and they are extracted from short-term and long-term LSTM models, respectively. The weights of all LSTM cells are initialized via Xavier [36] and are applied RNNDROP [37] with 0.7 keeping probability. In addition, Leaky ReLU [38] was used for activation function after FC layer and mean square error (MSE) is used for loss function. The adopted iterative optimization algorithm is stochastic gradient descent with the minibatch size of 64 using ADAM solver [39] and finally learning rate is set to 0.001. The training was with 5000 epochs for each model. Note that a basic form of the LSTM model is used as an implementation baseline, whereas the proposed framework is also applicable to more sophisticated LSTM modules since it is not much dependent on the structure of the internal LSTM module used.

B. Data Preprocessing

In this article, the proposed multiscale LSTM-based model with combinational hourly data was collected for a year from three different areas in South Korea, including Yeonseong, Sejong, and Yeongam. The data consist of historical PV output, slope solar irradiance, horizontal solar irradiance, panel surface temperature, and periphery temperature. Note that the proposed multiscale LSTM-based model aims to learn these meteorological and electric data so that VST-PVGF is feasible. The maximum rated voltages of PVGF for each area (Yeonseong, Sejong, and Yeongam) are 1490, 1625, and 3026 kW, respectively, and they are used for normalizing the predicted PV value. All the values of each field are normalized to $[-1, 1]$. After completing the normalization, the training data, as a composition of data samples from time step $t - d$ to $t - 1$, is utilized to predict PV output in time step t . Note that the size of training dataset

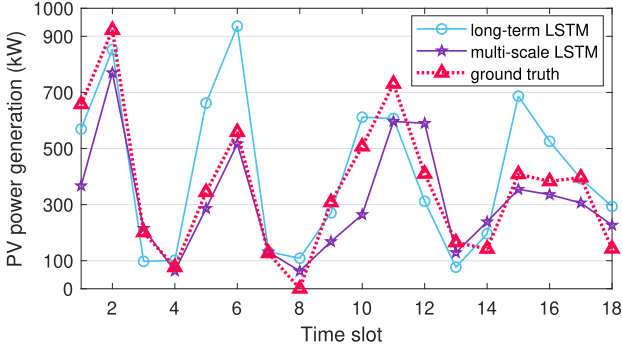


Fig. 5. Performance evaluation of a single model-based long-term LSTM and multiscale LSTM.

is nine months, and the size of test data set is three months. The d refers to the duration of training data. Compared to the previous short-term PV forecasting problem, which predicts daily PV output, hourly prediction through night and dawn, where PV power is not generated, is meaningless. This is because the amounts of PV generation during the period are close to zero so that the prediction for this duration is comparatively easy. Consequently, the overall prediction error can be reduced with this redundant data. Therefore, data between night and dawn is considered as *outlier* and is ignored for training the PV prediction model. After removing the redundant data, about 8000 training data was collected for each area. A single batch is combined with continuous 64 composed data each of which consists of continuous duration. The total batch group is shuffled for each other and 70% of an arbitrarily selected batch group among the total number of batches is utilized for training data and the other batch group is used as test data. For this regard, the continuity of sample data in each batch is maintained and it is useful to confirm whether the model reflects the PVGF temporal variation.

C. Performance Evaluation Metrics

For evaluating the proposed framework, the following metrics are used, i.e., RMSE, sMAPE [40], and MADE.

- 1) *RMSE*: It stands for the percentage of average sum of square errors between real and predicted values, and can be formulated as

$$\frac{100\%}{C} \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - y'_i)^2} \quad (1)$$

where C represents the maximum rated voltage of each area and y , y' , and n denote the measured ground truth, PV generation prediction amount, and the total number of test datasets, respectively.

- 2) *sMAPE*: It is exploited to distinguish the difference between two cases, which is difference between the ground truth and long-term LSTM and ground truth and multiscale LSTM in Fig. 5, i.e.,

$$\frac{100\%}{n} \sum_{i=1}^n \frac{|y_i - y'_i|}{|y_i| + |y'_i|}. \quad (2)$$

- 3) *MADE*: It is hard to select specific points when UUIT features occur to confirm the reactivity. As aforementioned, because the UUIT features affect the *solar ramping*, which reflects the reactivity, the effect of UUIT features can be measured as the metric of reactivity to evaluate the forecasting performance. Instead, the tendency of difference between ground truth and expected value can be used for estimating power generation under the consideration of UUIT features. However, traditional metrics such as sMAPE, RMSE, and nRMSE only consider absolute difference with respect to ground truth so that it is hard to confirm PVGF tendency. Thus, newly designed metric called MADE, which measures the error of the variation of forecast outputs is proposed as follows:

$$\text{MADE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{\Delta y_i - \Delta y'_i}{\Delta t} \right| \quad (3)$$

$$= \frac{1}{n} \sum_{i=1}^n |(y_i - y_{i-1}) - (y'_i - y'_{i-1})| \quad (4)$$

$$= \frac{1}{n} \sum_{i=1}^n |(y_i - y'_i) - (y_{i-1} - y'_{i-1})|. \quad (5)$$

Compared to other evaluation metrics, the sample data in the test dataset should be ordered in continuous time to evaluate the temporal variation. As previously mentioned, the temporal continuity can be maintained by combining the sample data in time order. Equation (3) can be reconstructed as (4) when the time interval is 1, and then it implies the derivative of the error function as formulated in (5). Even though the value of reactivity is higher for unexpected temporal fluctuation, MADE is not always lower if the accuracy for the common pattern is significantly inferior. Note that MADE is valid when two prediction models with the similar accuracy performance are compared, therefore, the MADE of short-term LSTM models, which have no ability to converge enough to forecast PV power generation, did not measured.

D. Experimental Results With Real-World Data

- 1) *Multiscale and single-scale model*: The proposed multiscale LSTM-based learning model is compared with single-model-based short-term, long-term LSTM models, and CNN model through experiments. Note that the data duration for the short-term and long-term LSTM model utilized 6 h and 24×7 h input data, respectively. In addition, the feature concatenation ratio of short LSTM and long LSTM in multiscale LSTM was set to 32 to 96. The single model-based LSTM consists of a single short-term [(g) in Table II] and long-term LSTM prediction model [(h) in Table II] showed 15.343 and 14.054 RMSE value, respectively. Moreover, the CNN-based forecasting model of (i) in Table II showed 25.925 RMSE value. However, the proposed multiscale LSTM-based model [(j) in Table II] showed 13.478 RMSE value, which is the lowest error value among compared models. In addition,

TABLE II
PERFORMANCE COMPARISON EXPERIMENTS BETWEEN SINGLE MODEL-BASED LSTM, SINGLE MODEL-BASED CNN, AND MULTISCALE LSTM MODEL FROM THREE AREAS: SEJONG, YEONGAM, AND YEONSEONG

Sejong (area #1)	Epoch	RMSE	sMAPE	MADE
single model-based short-term LSTM	1002	16.241	56.071	-
single model-based long-term LSTM	1405	14.531	49.963	178.061
multi-scale LSTM (ratio: 32 to 96)	1264	14.142	49.262	170.332
Yeongam (area #2)	Epoch	RMSE	sMAPE	MADE
single model-based short-term LSTM	1625	11.555	44.331	-
single model-based long-term LSTM	2551	10.014	34.172	177.416
multi-scale LSTM (ratio: 32 to 96)	2044	9.871	34.125	173.381
Yeonseong (area #3)	Epoch	RMSE	sMAPE	MADE
(g) single model-based short-term LSTM	1218	15.343	51.914	-
(h) single model-based long-term LSTM	1148	14.054	44.558	138.611
(i) single model-based short-term CNN	8000	25.925	80.965	254.195
(j) multi-scale LSTM (ratio: 32 to 96)	1075	13.478	42.008	133.508

TABLE III
EXPERIMENTS WITH RESPECT TO COUNTEREXAMPLE OF ENSEMBLE AND COMBINATION RATIO OF MULTISCALE LSTM-BASED MODEL

Results of each experiment	Epoch	RMSE	sMAPE	MADE
(k) short-term and short-term LSTM (ratio: 64 to 64)	1168	15.091	53.287	-
(l) long-term and long-term LSTM (ratio: 64 to 64)	1077	13.943	42.798	138.611
(m) long-term LSTM with 256 units	1046	13.773	42.954	137.508
(n) short-term and long-term LSTM (ratio: 16 to 112)	1121	13.818	43.099	135.227
(o) short-term and long-term LSTM (ratio: 32 to 96)	1075	13.478	42.008	133.508
(p) short-term and long-term LSTM (ratio: 64 to 64)	1318	14.038	44.394	138.372
(q) short-term and long-term LSTM (ratio: 96 to 32)	1030	15.017	51.703	164.697
(r) short-term and long-term LSTM (ratio: 112 to 16)	1061	15.117	51.457	168.839

The training data are based on Yeonseong.

as shown in Fig. 5, the proposed multiscale LSTM is close to the ground truth than long-term LSTM, with respect to *accuracy* and *reactivity* (because only the performance of single model-based short-term LSTM is inferior to compare with other models, it is not visualized in Fig. 5). It can be found in Table II, where the MADE and sMAPE of the multiscale LSTM are lowest than the others.

- 2) *Counterexample of ensemble*: Because validating the superior result of the multiscale LSTM-based model is not simply derived from ensemble or increased number of hidden LSTM units, corresponding experiments are conducted to compare several prediction models. The accuracy of the proposed multiscale LSTM-based model [(j) in Table II] is compared with LSTM with doubled units [(m) in Table III], concatenated model with short-term and short-term LSTMs [(k) in Table III], and long-term and long-term LSTMs [(l) in Table III]. The proposed model shows the improved prediction performance than other models. The proposed model showed 13.478 RMSE value, while other models [from (k) to (m) in Table III] showed 15.091, 13.943, and 13.773 RMSE value, respectively. In case of (m) in Table III, the prediction performance of the LSTM with doubled units is deteriorated compared to single model-based LSTM. This is because the model is overfitted to the given training data as the number of hidden units in its LSTM cell is excessively increased. Therefore, it confirms that the advantages of the concatenation approach of the proposed multiscale LSTM-based model is not simply derived from its ensemble-like structure. Furthermore, it has an additional advantage in terms

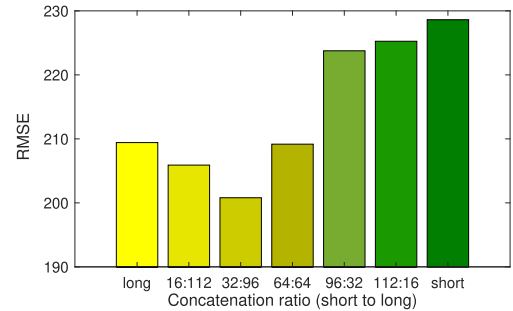


Fig. 6. Performance evaluation with the concatenation ratio variations between short and long-term LSTM models.

of the distribution of the backpropagation procedure to handle the gradient descent and overfitting issues.

- 3) *Combination ratio*: The last experiment that shows how much the ratio of each feature extracted from short-term and long-term LSTM in the multiscale model affects the prediction performance is conducted. To compare the performance of each ratio combination, five different experiments are conducted, as compared from (n) to (r) in Table III. As shown in Fig. 6, until converging to the optimal ratio, which is 32–96, the RMSE decreases as the ratio of short-term LSTM increases; however, thereafter the error increases again. It again confirms that the feature extracted individually from the bottleneck layers of short-term and long-term LSTMs have their own meaning and this framework is available to improve the performance for PVGF.

V. CONCLUSION AND FUTURE WORK

In this article, we propose an LSTM-RNN-based VST-PVGF framework to enable accurate and frequent estimation of PV power generation for smart city energy management. To improve reactivity while maintaining its accuracy, the proposed forecasting model exploits two different LSTM-based neural networks. To verify the validation and effectiveness of the proposed multiscale LSTM-based framework, several experiments are conducted. The results imply that 1) the reactivity of the multiscale LSTM is superior to others while maintaining its accuracy and 2) the proposed method is available to improve the overall performance of the baseline forecasting model. Throughout the data-intensive experiments, the proposed multiscale LSTM forecasting model in VST-PVGF problem outperforms the various forecasting models including baseline. As future work, further various performance evaluation with various models are desired in order to verify the performance of our proposed model.

In future research, more sophisticated models should be applied as baseline modules to verify the generality of the proposed multiscale learning framework. In addition, more scalable datasets can be collected with shorter intervals for more intensive experiments in terms of reactivity and accuracy. Finally, it would be worthy to utilize conventional statistics-based machine learning algorithms in order to presents the novelty of deep learning based algorithms.

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